



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BUILDING SETTING OUT

LSVBS302

Perform building set out

Competence

RQF Level: 3

Learning Hours



Credits: 6

Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land surveying

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Purpose statement	This module describes the skills, knowledge and attitudes required to perform building setting out. At the end of this module, participants will be able to select building set out tools and equipment, perform site preparation and building setting out.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Hygiene and safety precautions measures; Construction drawings interpretation; Linear distance measurement.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Select building set out tools and equipment	1.1.	Building set out tools are properly selected according to the type of the work.
	1.2.	Building setting out materials are efficiently selected according to the type of the work.
	1.3.	Building setting out equipment is correctly selected according to the type of the work.
2. Perform site	2.1.	Architectural drawings are properly analysed according to the needed information.

preparation	2.2. Level of the site is accurately checked according to the given reference datum.
	2.3. The site is properly cleared according to the types of waste.
3. Perform setting out of the building	3.1. Setting out methods are properly selected according to the types of work.
	3.2. Baseline and benchmark are accurately established according to architectural and layout drawings.
	3.3. Setting out method is properly applied according to the types of work.
	3.4. The working site is properly cleaned.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Select building set out tools and equipment 2. Perform site preparation 3. Perform setting out of the building
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Select building set out tools and equipment	Learning hours: 7
Indicative content	
* Surveying tools • Hammer, machete, Calibrated tape measure, tripod (ISO 12858-2), staff reading, ranging rod, hoe, spade, builder's square * Surveying materials • Building line, pegs, nails, timber board * Surveying equipment • Prism square, level, theodolite, PPE * Selection factors of surveying tool, materials, and equipment • Task to be accomplished • Accuracy required	

• Personal level of competence • Availability of equipment and tools	
Resources required for the learning outcome	
Equipment	• Level, theodolite, prism square, PPE. (RS ISO 20347, RS ISO 13688, ISO 3873)
Materials	• Pegs, building line, nails, pieces of chalk, pens / pencil , field note.
Tools	• Calibrated Tape measure, ranging rod, staff reading, ground mark, hammer, pins/arrows, plumb bob.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment, • Oral assessment, • Performance assessment, • Product based assessment

Perform site preparation	Learning hours: 13
Indicative content	
* Analyse Architectural drawing information • Scale, dimensions, foundation, wall, column, etc. * Perform levelling of the site • Operation of levelling instruments ✓ Temporal adjustment of level (Setting up, centering, levelling up, focusing eyepiece and objective) ✓ Permanent adjustment of level (Two peg test, plate bubble error, non-vertical crosshairs, optical plumb error, horizontal collimation error, trunnion axis dislevelment, vertical collimation error) • Determination of reduced levels (booking)	

✓ Rise and fall method ✓ Plane of collimation/height of instrument method • Determination of difference in height ✓ Spot height method • Describe baseline * Cleaning of the site • Removal of all unnecessary materials • Types of wastes materials • Methods of waste disposal	
Resources required for the indicative content	
Equipment	Compass, laser meter, level (ISO 17123-2), theodolite (ISO 8322-4)
Materials	Building line, nails, pencil ,pegs, timber, drawing documents ,batteries, etc
Tools	Calibrated Tape measure, spirit level, builder's square, ,etc
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment method	• Written assessment • Performance assessment • Product based assessment

Perform setting out of building		Learning hours: 40
Indicative content		
* Setting out methods (Refer to ISO 4463-2) • Optical method, • 3, 4, 5 rule or Pythagoras method, • Boulder's square method * Establish baseline and benchmark (Refer to ISO 4063 – 1) • Description building line, baseline, benchmark • Procedure to establish a baseline * Apply building setting out methods (Refer to ISO 4463-2) • Selection of the method • Procedure for the selected method • Demarcation of pegs * Cleaning of the working site (Refer to RS ISO 14015) ➤ Cleaning of the working area ➤ Cleaning of tools and equipment ○ Maintenance of safety and work tools and equipment ○ Safety precautions in cleaning tools and equipment ○ Safe store of equipment and tools		
Resources required for the indicative content		
Equipment	Level and its accessories (ISO 17123-2, ISO 1141) theodolite and its accessories(ISO 8322-4), PPE (RS ISO 20347, RS ISO 13688, ISO 3873),	
Materials	Pegs, timber board, building line(ISO 1969), nails(RS 364-1)	
Tools	Calibrated Tape measure, hammer (RS EAS 1021), spirit level (ISO 17123-2), hoe(RS EAS 137), spade (RS EAS 1020), wheelbarrow (RS 175), plumb bob, machete(RS EAS 188), pick ax	
Facilitation techniques	• Demonstration and simulation, • Individual and group work, • Practical exercise, • Trainer guided, • Group discussion	

Formative assessment methods	• Written assessment, • Performance assessment, • Product based assessment
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Integrated/Summative assessment

Integrated situation

ALTOS Engineering Company Ltd wants to build its office located in Kigali city, Gasabo district, Remera sector. As a surveyor of the company, you are requested to set out the building indicated in the detailed drawings and specifications provided by the company by using a dumpy level. Tools, materials, equipment, and the working site are provided.

- ✓ The building line is set to 6m from the center of the road
- ✓ The work should be done within 6 hours

Resources

Equipment	Level and its accessories, PPEs
Materials/ Consumables	Strings, pegs, timber board, nails, note book
Tools	Tape measure, Club hammer, claw hammer, spirit level, hoe, spade, wheelbarrow, plumb bob, machete, pick ax.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
1. Select building set out tools and	1.1 Building setting out tools are properly selected.	1. Hammer is selected.			2
		2. Tape measure is selected.			2
		3. Plumb bob is selected.			2

equipment (20%)		4. Spirit level is selected.			2
		5. Machete is selected.			2
	1.2 Building setting out materials are efficiently selected.	1. Pegs are selected.			2
		2. Strings are selected.			2
		3. Timber boards are selected.			2
	1.3 Building setting out equipment is correctly selected	1. Dumpy level is selected.			2
		3.PPEs are selected			2
2. Perform site preparation (20%)	2.2. Level of the site is accurately checked.	1. Setting up is well done.			2
		2. Centering is well done.			4
		3. Levelling is well done.			4
		4.Focusing is well done.			4
		5. Difference in levels is well calculated.			6
3. Perform setting out of the building (60%)	3.1. Setting out methods are properly selected.	1. Surveying instrument method is adopted.			4
	3.2. Accurate establishment of baseline and benchmark.	1. Baseline is well established.			4
		2.Benchmark is well established.			4
	3.3. Setting out methods are properly applied.	1. Dimension is well respected.			10
		2. Angles are well measured.			6

		3. External corners are well established.			5
		4. size of the trench is well established.			5
		5. Size of the foundation is well established.			4
		6. Size of the wall is well established.			5
		7. Diagonals are well checked.			5
	3.4 The working site is properly cleaned.	1. Tools and equipment are well cleaned.			4
		2. Working area is well cleaned.			4
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. Charles, D., Ghilani, P. (2012). Elementary surveying 13th edition: Pennsylvania State. Prakasam, Pune
2. Kanetkar, T. P., and Kulkarni, S. V. (1981). Surveying and Leveling, Vol I. Pune Vidyarthi Griha: Kavanagh, B (2009). Surveying Principle and Application 8th edition.
3. Oregon Department of Transportation Geometronics Unit (2000). Basic surveying theory theory and practice: Bend, Oregon
4. Schofield, W and Breach, M. (2007). Engineering surveying 6th edition: Kingston University
5. ISO 8322-4 Building construction — measuring instruments — Procedures for determining accuracy in use — Part 4: Theodolites
6. ISO 17123-2 Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 2: Levels
7. RS ISO 14015: Environmental management systems — Environment Assessment of Sites and Organizations
8. ISO 4463-2 Measurement methods for building — setting-out and measurement — part 2: measuring stations and targets
9. ISO 4463-1 Measurement methods for building — setting-out and measurement — part 1: planning and organization, measuring procedures, acceptance criteria
10. R.S ISO 20347 Personal protective equipment — Occupational footwear
11. RS ISO 13688 /Amd 1:2021 protective clothing — general requirements — amendment 1
12. ISO 3873 Industrial safety helmets
13. RS 364 nails — specification — part 1: steel nails
14. RS EAS 1021 steelhead hammer — specification
15. RS EAS 188 machete — specification
16. ISO 1969 Fibre ropes — Polyethylene — 3- and 4-strand ropes
17. ISO 1141 Fibre ropes — Polyester — 3-, 4-, 8- and 12-strand ropes
18. ISO 12858-2:2020 optics and optical instruments — ancillary devices for geodetic instruments — part 2: tripods